VOLUME 7 — MODEL PAPER 2 (Difficulty: Moderate)

NID DAT (B.Des) Prelims: Practice Model Paper 2
Handbook-scheme format. This practice paper was written for this package; it is not an official NID paper.
Focus mix: observation, spatial reasoning, Indian crafts, pictograms, elevated-viewpoint scene, memory + writing, sustainable toy design.

INSTRUCTIONS

1. Time: **3 hours**. Total: **100 marks**. No negative marking.
2. Part I: Q1–14 (1 mark each), Q15–22 (2 marks each) = 30 marks.
3. Part II: Q23 (15), Q24 (20), Q25 (15), Q26 (20) = 70 marks.
4. Suggested timing: Part I = 25 min · Q23 = 25 · Q24 = 40 · Q25 = 28 · Q26 = 40 · buffer = 22.

PART I — OBJECTIVE (30 marks)

Q1–Q14 (1 mark each)

Q1. Which capital letter has rotational symmetry of order 2 but NO line of symmetry?
(a) A (b) Z (c) E (d) T

Q2. Pochampally Ikat is woven in:
(a) Odisha (b) Telangana (c) Gujarat (d) Assam

Q3. Which of these is NOT a primary colour of light (RGB)?
(a) Red (b) Green (c) Blue (d) Yellow

Q4. Which of these is a *lettermark* logo?
(a) Apple (b) IBM (c) Nike swoosh (d) Starbucks siren

Q5. An arrow points East. It is rotated 270° clockwise. Where does it point now?
(a) North (b) South (c) West (d) East

Q6. Which Indian craft uses quartz powder instead of clay?
(a) Khurja pottery (b) Jaipur Blue Pottery (c) Longpi pottery (d) Bankura terracotta

Q7. Loom : Weaver :: Kiln : ?
(a) Blacksmith (b) Potter (c) Carpenter (d) Tailor

Q8. How many lines of symmetry does a square have?
(a) 2 (b) 4 (c) 6 (d) 8

Q9. "Less is more" is a motto closely associated with:
(a) Baroque (b) Modernism (c) Rococo (d) Memphis Design

Q10. Odd one out: Gond, Warli, Madhubani, Chikankari
(a) Gond (b) Warli (c) Madhubani (d) Chikankari

Q11. How many A4 sheets have the same area as one A3 sheet?
(a) 1.5 (b) 2 (c) 3 (d) 4

Q12. Which word looks unchanged in a water image (flipped top to bottom)?
(a) BOX (b) CAT (c) TOP (d) MAN

Q13. At 5 pm, in which direction does your shadow point?
(a) North (b) South (c) East (d) West

Q14. The handle of a cup "inviting" you to hold it is an example of:
(a) Feedback (b) Affordance (c) Mapping (d) Constraint

Q15–Q22 (2 marks each)

Q15. How many squares of all sizes are there in a grid of 3 rows × 4 columns of unit squares?
(a) 12 (b) 18 (c) 20 (d) 24

Q16. Statements: All bowls are vessels. Some vessels are made of brass.
Conclusions: I. Some bowls are made of brass. II. Some vessels are bowls.
(a) Only I follows (b) Only II follows (c) Both follow (d) Neither follows

Q17. An 8 cm × 8 cm sheet is folded so the bottom half goes up onto the top half (fold at y = 4). Then the left half is folded onto the right half (fold at x = 4). A hole is punched at (6, 6). Where are the holes when the sheet is unfolded? (x is measured from the left edge, y from the bottom edge.)
(a) (6,6) and (2,6)
(b) (6,6), (2,6), (6,2), (2,2)
(c) (6,6), (6,2)
(d) (2,2) only

Q18. A standard die (opposite faces add up to 7) shows 1 on top, 2 in front and 3 on the right. Which number is on the left face?
(a) 4 (b) 5 (c) 6 (d) 2

Q19. On a 1 : 50 drawing, a bed measures 4 cm × 3.8 cm. What is its real floor area?
(a) 1.9 m² (b) 3.8 m² (c) 7.6 m² (d) 15.2 m²

Q20. A village school has no electricity, and its classrooms become very hot in summer. Which is the BEST low-cost design approach?
(a) Install air-conditioners powered by a diesel generator
(b) Use a white reflective roof coating, cross-ventilation openings, jaali screens and shade trees
(c) Close all windows to keep the heat out
(d) Paint the walls dark brown

Q21. Dots arranged as ●● ●● ●● (pairs with gaps between them) are seen as three groups. Which Gestalt principle explains this?
(a) Similarity (b) Proximity (c) Closure (d) Continuity

Q22. A person walks 4 km North, then 3 km East, then 4 km South. How far are they from the starting point, and in which direction?
(a) 3 km East (b) 3 km West (c) 5 km North-East (d) 11 km East

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PART II — SUBJECTIVE (70 marks)

Q23. Pictogram System (15 marks)
Design **four pictograms** for a public park: (i) drinking water, (ii) toilet, (iii) "do not pluck flowers", (iv) dustbin. They must look like **one family** (consistent style). Draw each inside a 6 cm × 6 cm square.

Q24. Scene in Perspective (20 marks)
Draw a **Sunday-morning vegetable market** as seen by a person **standing on the back of a parked truck** (a slightly elevated view). Include at least **eight people** doing different things.

Q25. Memory + Writing (15 marks)
(a) Draw **your school bag from memory** in a 3/4 view (8 marks).
(b) In about **120 words**, describe **one improvement** you would make to it, and why (7 marks).

Q26. Sustainable Toy Design (20 marks)
Design a **toy for a five-year-old child** using **only waste materials** commonly found at home. Show: (a) the toy (main drawing), (b) the materials used (labelled), (c) how a child plays with it (a small scene), and (d) one safety consideration.

ANSWER KEY — PART I

Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	b	7	b	13	c	19	b
2	b	8	b	14	b	20	b
3	d	9	b	15	c	21	b
4	b	10	d	16	b	22	a
5	a	11	b	17	b		
6	b	12	a	18	a		

DETAILED SOLUTIONS — PART I

Q1 → (b) Z. Z looks the same after a half turn but has no mirror axis. A and T have a vertical axis; E has a horizontal one. *Concept:* symmetry types · *Diff:* M.

Q2 → (b) Telangana. The weave is named after Pochampally (Bhoodan Pochampally) in Nalgonda/Yadadri district; it holds a GI tag. *Concept:* textiles · *Diff:* E.

Q3 → (d) Yellow. The light primaries are red, green and blue. Yellow is a painter's (RYB) primary, and it is also what red + green light makes. *Concept:* colour models · *Diff:* E.

Q4 → (b) IBM. It is made of initials (Paul Rand's striped version). Apple is pictorial; the swoosh is abstract; the siren is an emblem or mascot. *Concept:* logo types · *Diff:* E.

Q5 → (a) North. 90° clockwise from East = South, 180° = West, 270° = North. Equivalently, 90° anticlockwise from East is North. *Concept:* rotation · *Diff:* E.

Q6 → (b) Jaipur Blue Pottery. It uses quartz stone powder, powdered glass and Multani mitti, not clay. *Concept:* materials · *Diff:* M.

Q7 → (b) Potter. This is the main equipment–worker pair: a weaver uses a loom, a potter fires in a kiln. *Concept:* analogy · *Diff:* E.

Q8 → (b) 4. Two diagonals + two midlines. *Concept:* symmetry · *Diff:* E.

Q9 → (b) Modernism. The phrase is associated with Mies van der Rohe. *Concept:* design history · *Diff:* E.

Q10 → (d) Chikankari. It is embroidery (Lucknow); the others are painting traditions. *Concept:* classification · *Diff:* E.

Q11 → (b) 2. Each A-size is half of the size before it. *Concept:* paper sizes · *Diff:* E.

Q12 → (a) BOX. B, O and X are all symmetric about a horizontal axis. In CAT, the A and T are not. In TOP, the T and P are not. In MAN, the M, A and N are not. *Concept:* water image · *Diff:* M.

Q13 → (c) East. The evening sun is in the west, so shadows point east. *Concept:* shadows · *Diff:* E.

Q14 → (b) Affordance. The handle's form suggests the action of holding. *Concept:* HCD · *Diff:* E.

Q15 → (c) 20. 1×1 squares: 3 × 4 = 12. 2×2 squares: 2 × 3 = 6. 3×3 squares: 1 × 2 = 2. Total 20. *Concept:* counting · *Diff:* M.

Q16 → (b) Only II. "All bowls are vessels" guarantees that some vessels are bowls (II). The brass vessels may or may not include bowls, so I is only possible, not certain. *Concept:* syllogism · *Diff:* M.

Q17 → (b). Unfold in reverse order. First undo the left-onto-right fold (x = 4): (6,6) reflects to (2,6). Then undo the bottom-onto-top fold (y = 4): (6,6) reflects to (6,2) and (2,6) reflects to (2,2). Four holes: (6,6), (2,6), (6,2), (2,2). *Concept:* folding · *Diff:* M.

Q18 → (a) 4. The left face is opposite the right face (3), and 7 − 3 = 4. *Concept:* dice · *Diff:* E.

Q19 → (b) 3.8 m². Real size: 4 × 50 = 200 cm = 2 m, and 3.8 × 50 = 190 cm = 1.9 m. Area = 2 × 1.9 = 3.8 m². *Concept:* scale and area · *Diff:* M.

Q20 → (b). It needs no electricity and uses passive cooling: reflection, ventilation, shading and evaporative tree cover. (a) is costly and polluting, (c) traps heat, and (d) absorbs more heat. *Concept:* sustainable, context-appropriate design · *Diff:* M.

Q21 → (b) Proximity. Elements close together are seen as a group. *Concept:* gestalt · *Diff:* E.

Q22 → (a) 3 km East. The north and south legs cancel out, leaving the 3 km east leg. *Concept:* directions · *Diff:* E.

DETAILED SOLUTIONS — PART II

Q23. Park pictograms (15 marks)

Model system

- **Grid:** a 6 cm square with a 0.5 cm margin. All figures use the same stroke weight (filled silhouettes) and the same "human" figure style: a round head and a rounded-rectangle body.
- **Colour code:**
  - blue background + white symbol = information (water, toilet, dustbin)
  - red circle and slash on white = prohibition (do not pluck flowers)
- **Designs:**

1. *Drinking water:* a tap with three droplets falling into a cup, or a person drinking from a fountain spout.
2. *Toilet:* standard male and female figures side by side with a vertical divider (a widely understood convention); an accessible-toilet variant can be added.
3. *Do not pluck flowers:* a hand reaching for a flower, with a red circle and diagonal slash over it.
4. *Dustbin:* a figure dropping an item into a bin (more active, and teaches the behaviour).

Rubric (15)

Criterion	Marks
Instant recognisability of each icon	5
Consistency (stroke, figure style, grid, colour code)	4
Simplicity and no unnecessary text	3
Correct use of prohibition convention	2
Neat execution	1

Mistakes: a different figure style in each icon; using words instead of symbols; overly detailed drawings.

Q24. Sunday vegetable market from a truck (20 marks)

Approach

- **Viewpoint:** about 2.5–3 m high, so the **horizon line sits high in the frame**. We look down on the tops of umbrellas and baskets, and nearby people show the tops of their heads. Use 2-point perspective with slightly converging verticals (a hint of a 3rd VP below), or simplified 2-point.
- **Layout:** the foreground shows the edge of the truck bed (frames the view, explains the viewpoint). The middle ground has rows of stalls with tarpaulin umbrellas, mounds of tomatoes, onions and greens, weighing scales and baskets. The background has buildings, a temple top, and wires.
- **8+ people with different actions:**
  - a vendor weighing tomatoes
  - a woman bargaining
  - a child pointing at mangoes
  - an old man with a cloth bag
  - a porter carrying a sack on his head
  - a boy on a cycle
  - a customer paying with a phone (UPI QR code on the stall)
  - a cow eating discarded leaves
  - a delivery person
- **Colour:** fresh produce colours make a natural focal point. Use warm morning light, with shadows falling to one side.

Rubric (20)

Criterion	Marks
Correct elevated viewpoint (high horizon, top views)	5
People: number, variety of actions, proportion	5
Observation of authentic market details	4
Composition & depth layers	3
Rendering / colour	3

Mistakes: drawing an eye-level view (ignoring "on a truck"); people all facing the viewer; no produce detail.

Q25. School bag + 120 words (15 marks)

(a) Drawing (8): a 3/4 view showing the main compartment, front pocket, side bottle pocket, padded straps, top handle, zippers with pullers, a brand patch and reflective strip, and the bottom panel. Add wear marks for realism (an observation bonus).

(b) Model 120-word answer:

> My school bag's biggest problem is weight distribution. All my heavy textbooks sit at the back of a single deep compartment, so the bag pulls away from my shoulders and I lean forward while walking. I would add a padded internal divider with an elastic strap to hold heavy books close to my back, plus a detachable waist belt that transfers some weight to the hips. A firm back panel with air channels would reduce sweating on hot afternoons. The side bottle pocket should also be deeper, because my bottle falls out on the bus. These changes cost little but would protect my posture every day, which matters because I carry this bag for over 200 days a year.

(About 120 words.) Structure: problem → cause → solution(s) → benefit.

Rubric: drawing accuracy 4 + detail 2 + rendering 2; writing relevance 3 + clarity 2 + word-limit discipline 2.

Q26. Toy from waste (20 marks)

Model concept: "Bottle-Cap Caterpillar Pull-Toy"

- **Materials:**
  - 6 plastic bottle caps (washed)
  - an old shoelace or string
  - a cardboard tube (head)
  - a shoebox lid cut into wheels
  - an old sock (the caterpillar's body cover)
  - buttons (eyes)
  - bottle-cap wheels on toothpick axles
- **Main drawing:** a 3/4 view of the caterpillar made of linked segments on cap wheels, with a pull string.
- **Exploded labels:** segment = cardboard ring + sock cover; wheels = caps; axle = ice-cream stick; joints = shoelace threaded through holes.
- **Play scene:** a 5-year-old pulls it across the floor; the segments wiggle. A second use: stack the caps to learn colours and counting.
- **Safety:** no small detachable parts for children under 3 (secure the buttons by stitching); round the edges; use non-toxic glue; no sharp toothpick ends (cover them with cap bits).

Rubric (20)

Criterion	Marks
Uses only household waste; materials clearly labelled	4
Play value for the age group (fun, learning)	4
Clarity of main drawing + construction	5

Criterion	Marks
Play scene communicates use	3
Safety consideration	2
Originality	2

Mistakes: materials that aren't household waste (e.g. bought motors); ignoring safety for young children; no play scene.

SCORING INTERPRETATION — MODEL PAPER 2

Practice benchmarks only, not NID cutoffs.

Score	Band
78–100	**Excellent**
62–77	**Strong**
48–61	**Good**
33–47	**Needs improvement**
< 33	**Major improvement required**

Diagnostic tip: if Q24 scored low because of the viewpoint, practise 5 elevated-view sketches (from a balcony, bridge or bus top) this week.

NEXT VOLUME: Volume 8 — Model Paper 3 (`08-VOL8-MODEL-PAPER-3.md`)